Calibrating the moist Sobel–Schneider model against ERA5: what the reanalysis settles, and what it does not

2026-08-01

Summary

The model's moisture, thermodynamic and momentum equations all contain a velocity written v, and they do not mean the same velocity by it. The moisture equation's $-(2a-1)vW$ needs two branches that between them exhaust the column, so each holds half the column mass. The thermodynamic equation's $(d\Delta_z/H)\partial_y v$ describes a layer of depth d at the top of the troposphere and another at the bottom, with $v = 0$ in between. Those two velocities differ by the ratio of their depths, a factor of about 2.5 at the observed surface pressure, and every coefficient defined as "transport per unit v" scales with that factor. Measuring one coefficient under one reading and pairing it with another measured under the other reading is what produced the negative gross moist stability that has been the obstacle since 2026-07-31. It was an accounting error, not a physical result.

Two combinations of the coefficients do not depend on the reading at all: the ratio $\hat{H}/\hat{S}$, and the column $W^* = \hat{S}/(L_v(2a-1))$ at which the gross moist stability changes sign. Those are the right things to compare with observations, and they can be compared without ever deciding what v means. ERA5 gives W^* in the range 44 to 57 $\mathrm{kg\,m^{-2}}$, depending on how the time mean is taken, against 44.3 $\mathrm{kg\,m^{-2}}$ for the model at its default $a = 0.85$. The model's default therefore sits at the extreme strong-transport end of what the observations allow, and it is the one choice that puts a $W_c = 50$ run below its own sign change, since such a run settles at a quiescent column of 50.7 $\mathrm{kg\,m^{-2}}$.

Acting on that: a V2 run at $W_c = 50$ with a lowered to 0.77 was integrated to convergence for this report. The aggregated regime of 2026-07-31, five precipitating bands, a 78 $\mathrm{mm\,day^{-1}}$ equatorial spike and a bone-dry third of the domain, does not appear. The run instead reproduces the recognizable single-ITCZ circulation of the $W_c = 40$ control almost exactly (Fig. 8).

The model's gross dry stability, which I reported on 2026-07-31 as 2.55 times too small, is not. That claim came from dividing the observed dry-static-energy flux by a half-column velocity while comparing against a layer-velocity $\hat{S}$. Read consistently, the model's $\hat{S} = Cd\Delta_z/H = 7.75 \times 10^7$ $\mathrm{J\,m^{-2}}$ corresponds to a slab pair 195 hPa deep, which the observed $[v]$ profile accommodates comfortably. Measured against the branch depth the observed wind shape itself implies, 238 hPa, the model's $\hat{S}$ is 18% low. Both that claim and the 2026-07-31 recommendation of $a = 0.95$ are withdrawn.

What ERA5 does not settle is the sign of the model's gross moist stability at its current defaults. Four defensible ways of taking the time mean give a between 0.77 and 0.84 and W^* between 44

and 57 $\mathrm{kg\,m^{-2}}$, a spread that straddles the quiescent column. Section 7 lays that out; Section 13 lists what would narrow it.

1 What the model asks of the observations

Three terms in the model transport something meridionally, and all three are written in terms of one velocity v (Appendix A has the full equations):

$$\text{moisture:} \quad \partial_y[-(2a-1)\,v\,W]\,, \tag{1}$$

$$\text{thermodynamic:} \quad \frac{d\,\Delta_z}{H}\,\partial_y v, \tag{2}$$

$$\text{momentum:} \quad \text{the factor of 2 in the } v \text{ equation.} \tag{3}$$

Equation (1) is built from a two-branch column. A fraction a of the column water vapor W sits in the lower branch and moves at $-v$; the remaining fraction $1-a$ sits in the upper branch and moves at $+v$. The column-integrated flux is then $(1-a)W(+v)+aW(-v) = -(2a-1)vW$. For the fractions to be a and $1-a$ the two branches must exhaust the column between them. Since they also have equal and opposite velocities, and time-mean mass balance requires their mass transports to be equal and opposite, they must have equal masses. Two branches of equal mass that exhaust the column each hold half of it, so the interface is at $p_s/2$ and the branch velocity is

$$v_{\text{half}} = \frac{2g\,\Psi_{\text{ext}}}{p_s}, \tag{4}$$

with Ψ_{ext} the observed overturning mass transport, that is, the zonal-mean mass streamfunction evaluated at the level where $[v]$ reverses sign.

Equation (2) is built from a different column. It comes from vertical advection $w\,\partial_z\theta$ with $w = d\,\partial_y v$, the vertical velocity that continuity gives when the horizontal flow occupies a layer of depth d, and $\partial_z\theta = \Delta_z/H$. That picture has a moving layer of depth d at the top, another at the bottom, and $v = 0$ between them. A slab of pressure depth dp carrying mass transport Ψ_{ext} moves at

$$v_\delta = \frac{g\,\Psi_{\text{ext}}}{dp}. \tag{5}$$

Dividing (5) by (4),

$$\frac{v_\delta}{v_{\text{half}}} = \frac{p_s/2}{dp}. \tag{6}$$

At the observed tropical $p_s = 997$ hPa and a slab depth near 195 hPa this ratio is 2.55. Any coefficient defined as a transport divided by v inherits it exactly. That single factor is the whole of the discrepancy I reported on 2026-07-31.

2 The part of the answer that does not depend on the reading

Before choosing between the two readings, it is worth isolating what does not require the choice. The gross moist stability is

$$\hat{H} = \hat{S} - L_v(2a-1)W, \tag{7}$$

and both $\hat{S}$ and $(2a-1)$ are transports per unit v, so both scale with the same factor (6). Two combinations therefore do not scale at all:

$$W^* \equiv \frac{\hat{S}}{L_v(2a-1)} = \frac{\hat{S}\,W}{\hat{S}-\hat{H}}, \qquad \frac{\hat{H}}{\hat{S}} = 1 - \frac{W}{W^*}. \tag{8}$$

W^* is the column at which $\hat{H}$ changes sign. In V1 and V2 the equilibrium column is nearly uniform at the quiescent value $W_q = W_c + \tau_c E_0$ set by the local evaporation–precipitation balance, so the comparison of W_q with W^* alone decides which regime a run is in. The second form of W^* in (8) needs only the two regressed stabilities and never names a value of $2a-1$, so it cannot import a normalization along with it.

Measured over $|\varphi| \leq 20°$ from ERA5 2000–2019, with the barotropic mass adjustment applied to $[v]$ (Section 8):

$$W^* = 56.8\ \mathrm{kg\,m^{-2}}, \qquad \frac{\hat{H}}{\hat{S}} = 0.303 \quad \text{at } W = 39.6\ \mathrm{kg\,m^{-2}}. \tag{9}$$

Table 1 puts the model beside that. At the model's fixed $\hat{S}$, its a is the only free dial in W^*, and the default $a = 0.85$ puts the sign change at 44.3 $\mathrm{kg\,m^{-2}}$, below the 50.7 $\mathrm{kg\,m^{-2}}$ quiescent column that the V1 defaults produce. Every V2 run at $W_c = 50$ was therefore in the negative-stability corner before it started.

Table 1: The two normalization-free comparisons. $\hat{H}/\hat{S}$ is evaluated at the quiescent column $W_q = 50.7\ \mathrm{kg\,m^{-2}}$ that the V1 defaults produce, which is the value the model actually sits at. The ERA5 row uses the transport-weighted regression over every month; Section 7 gives the spread across other time means.

	a	$2a-1$	W^* $(\mathrm{kg\,m^{-2}})$	$\hat{H}/\hat{S}$ at W_q
model default	0.850	0.700	44.3	−0.145
ERA5-consistent	0.773	0.546	56.8	+0.107

3 The d/H argument, laid out

This is the step that was unclear in chat, so it is written out slowly here.

Step 1. The thermodynamic equation fixes the dry-static-energy flux; it is not a separate assumption. The model's temperature equation contains $(d\Delta_z/H)\partial_y v$. Multiplying by the column heat capacity $C = c_p \Pi\, \Delta p/g$ turns that term into the divergence of a column dry-static-energy flux, so

$$F_{\mathrm{DSE}} = C\,\frac{d\,\Delta_z}{H}\,v \equiv \hat{S}v, \qquad \hat{S} = C\,\frac{d\,\Delta_z}{H}. \tag{10}$$

With $C = 5.164 \times 10^6\ \mathrm{J\,m^{-2}\,K^{-1}}$, $d = 4$ km, $\Delta_z = 60$ K and $H = 16$ km, $\hat{S} = 7.746 \times 10^7\ \mathrm{J\,m^{-2}}$. Nothing has been assumed beyond the equation itself.

Step 2. $d\Delta_z/H = 15$ **K is a temperature difference across one layer of depth** d**.** That number looks small next to $\Delta_z = 60$ K, and that is what made the model's $\hat{S}$ look wrong. But it is only half the story, because $\hat{S}$ also contains the full-column C.

Step 3. Whether 15 K is right depends entirely on what v means. Write the same flux as a two-slab transport: mass transport Ψ_{ext} times the dry-static-energy difference Δs between the branches. With $\Psi_{\text{ext}} = v\,dp/g$,

$$F_{\text{DSE}} = \frac{v\,dp}{g}\,\Delta s \quad\Longrightarrow\quad \hat{S} = \frac{dp\,\Delta s}{g} \quad\Longrightarrow\quad \Delta s = \frac{g\,\hat{S}}{dp}. \tag{11}$$

The model's $\hat{S}$ is one number; the branch temperature difference it implies is whatever (11) gives for the slab depth in question. Equating $\Delta s = c_p \Pi\,\Delta\Theta$ and using $\Delta p = p_s - p_t$, $\Delta\Theta = (\Delta p/dp)(d\Delta_z/H)$. At $dp = \Delta p/2$, that is 30 K. At $dp = 195$ hPa, it is 62 K, comparable to Δ_z itself. The "15 K" is a per-layer number that only becomes a branch contrast once a layer depth is named.

Step 4. The observed Δs settles it. ERA5's dry-static-energy flux per unit overturning is $F_{\text{DSE}}/\Psi_{\text{ext}} = 38.9\ \text{kJ}\,\text{kg}^{-1}$ over $|\varphi| \leq 20°$. Putting that into (11) with the model's $\hat{S}$ gives $dp = 195$ hPa. So the model's gross dry stability is exactly the statement that its two branches are slabs 195 hPa deep, $39\ \text{kJ}\,\text{kg}^{-1}$ apart in s.

Step 5. The alternative I floated in chat was wrong. I suggested that the thermodynamic equation should carry $(H - d)\Delta_z/H = 45$ K instead of 15 K. That reproduces the literal-bulk half-column $\hat{S}$ to 2% (2.324×10^8 against 2.283×10^8), which is why it looked right. It is the wrong fix, because it repairs a layer-velocity equation by matching a half-column measurement. The two-readings table of the next section is the correct picture.

4 The two readings, measured

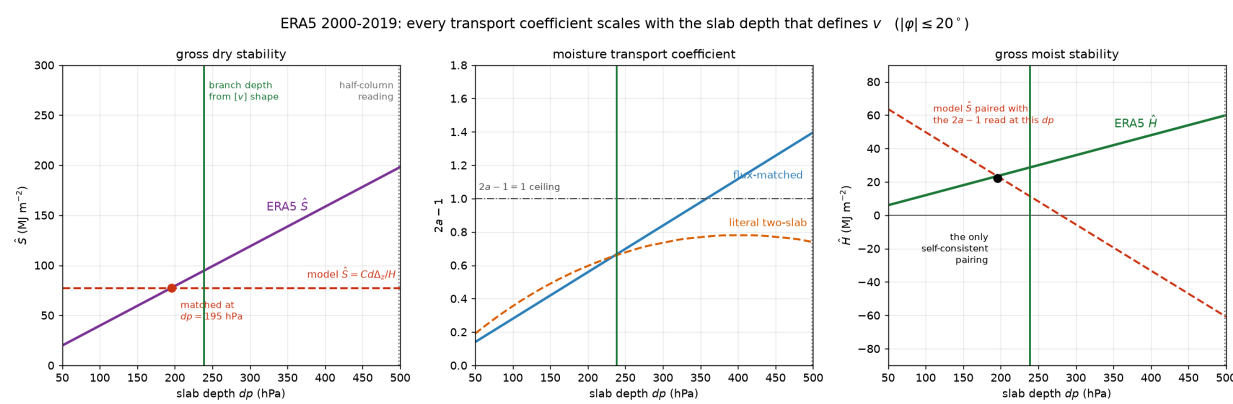

Figure 1: Every transport coefficient against the slab depth that defines v. Purple, blue and green are ERA5. Red dashed in the left panel is the model's $\hat{S} = Cd\Delta_z/H$, a fixed number, whose intersection with the observed curve at 195 hPa is the depth the model's dry stability corresponds to. Green vertical line: the branch depth the observed $[v]$ profile implies on its own, 238 hPa. Grey dotted: the half-column reading, $dp = p_s/2$. Right panel, red dashed: what $\hat{H}$ becomes if the model's $\hat{S}$ is paired with a $2a-1$ measured at some other depth, which is the mixing that produced the negative gross moist stability; the black dot at 195 hPa is the only self-consistent pairing. Centre panel, orange: what $2a-1$ would be for two genuinely uniform slabs of that depth, against the blue coefficient the observed flux actually demands.

Table 2 is Fig. 1 in numbers. Every column is linear in dp, as (6) requires, and W^* is constant down the table, as (8) requires. The half-column row is the $dp = 499$ hPa member of the same family, and its $2a - 1 = 1.394$ exceeds the ceiling $2a - 1 \le 1$ that follows from $a \le 1$: a signal, visible at the time, that the half-column reading was not the model's.

Table 2: Transport coefficients under each reading of v. ERA5 2000–2019, $|\varphi| \le 20°$, mass-adjusted $[v]$, regressed over all months. Observed tropical $p_s = 997$ hPa and $W = 39.6\ \mathrm{kg\,m^{-2}}$.

reading	$\hat{S}$ ($\mathrm{J\,m^{-2}}$)	$2a - 1$	a	$\hat{H}$ ($\mathrm{J\,m^{-2}}$)	W^* ($\mathrm{kg\,m^{-2}}$)
half-column, $dp = p_s/2$	1.978×10^8	1.394	1.197	5.99×10^7	56.7
slab, $dp = 150$ hPa	5.943×10^7	0.419	0.709	1.80×10^7	56.8
slab, $dp = 196$ hPa	7.766×10^7	0.547	0.774	2.35×10^7	56.8
slab, $dp = 250$ hPa	9.906×10^7	0.698	0.849	3.00×10^7	56.8
slab, $dp = 300$ hPa	1.189×10^8	0.838	0.919	3.60×10^7	56.8

Mixing the readings is what produced the negative gross moist stability. Holding the model's layer-velocity $\hat{S}$ fixed and pairing it with each candidate coefficient:

combination	$\hat{H}$ ($\mathrm{J\,m^{-2}}$)
model $\hat{S}$ with the half-column flux-matched $2a - 1 = 1.394$	-6.07×10^7
model $\hat{S}$ with the half-column mass partition $2a - 1 = 0.900$	-1.17×10^7
model $\hat{S}$ with the slab $2a - 1 = 0.546$ at the matched depth	$+2.34 \times 10^7$

5 Equal pressure depths, and how thick the branches really are

Spencer's correction of 2026-08-01 was that the two layers must have equal depths in *pressure*, not equal geometric depths, and it is what makes the slab picture self-consistent. Equal dp gives equal mass, and equal mass is what equal and opposite branch velocities require. Measured in ERA5, a 4 km layer spans 153 hPa just below the model's tropopause and 367 hPa just above the ground in the $\pm 12°$ composite of Fig. 2 (369 hPa averaged over $|\varphi| \le 20°$), a mass ratio of 2.4, so the model's own geometry read literally cannot give its two layers equal mass. Fig. 2 panel (a) draws both readings on the observed $[v]$ profile.

Three routes lead to a slab depth, and they do not agree:

route	depth
from the observed $s(p)$ profile, matching $\Delta s = 38.9\ \mathrm{kJ\,kg^{-1}}$	90 hPa
from the model's own $\hat{S}$, through (11)	195 hPa
from the shape of the observed $[v]$, with no energy data used	238 hPa

The third is the one that makes the comparison a test rather than a fit. For each branch of the annual-mean $[v]$, split at the level of nondivergence, the participation ratio $\left(\int |v|\,dp\right)^2 / \int v^2\,dp$ returns the depth of the top hat that carries the same transport at the same peak-equivalent speed, and it uses only the wind. Averaged over 5°–25° it gives 296 hPa for the upper branch, 180 hPa for the lower, 238 hPa in the mean. Against the observed $\hat{S}$ at that depth, $9.44 \times 10^7\ \mathrm{J\,m^{-2}}$, the model's $\hat{S}$ is 18% low.

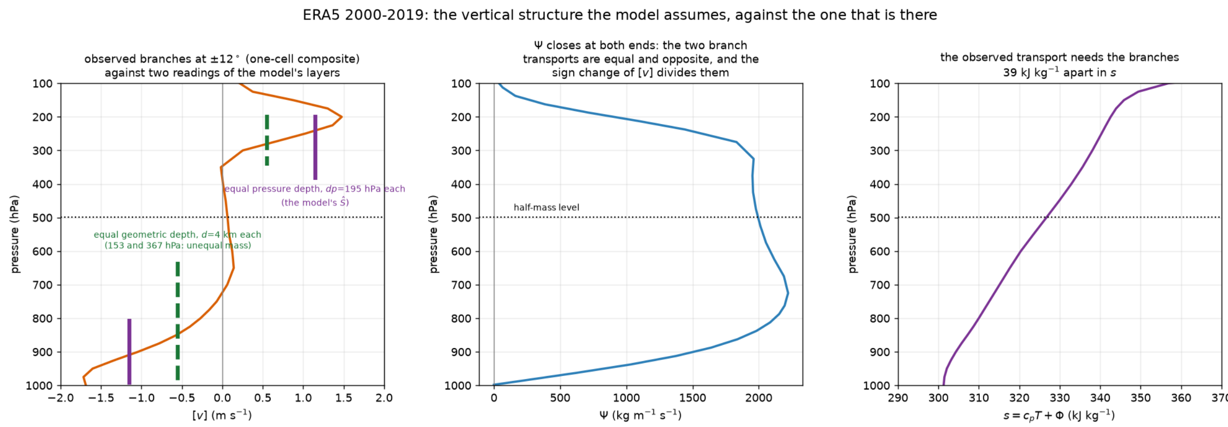

Figure 2: (a) The observed $[v]$ at $\pm 12°$, composited into one cell, with the model's two layers drawn two ways: equal geometric depth of 4 km each (green dashed, 153 and 367 hPa, unequal mass) and equal pressure depth of 195 hPa each (purple, the depth the model's $\hat{S}$ implies). (b) Ψ vanishes at both ends once the column mass flux is removed, which is the statement that the two branch transports are equal and opposite; the level where $[v]$ changes sign is what divides them. (c) The observed $s(p)$, whose branch contrast the transport must reproduce.

The first route disagreeing with the others is the honest residual of the two-slab idealization. Two 195 hPa slabs taken from the observed $s(p)$ profile differ by 33.7 $\mathrm{kJ\,kg^{-1}}$, 13% short of the 38.9 the transport needs, because the real branches correlate $[v]$ with s inside themselves and a uniform slab cannot. That is a real limitation on how precisely d can be pinned, and it is a 13% effect rather than the factor of 2.5 that mixing normalizations produced.

One consequence for branch speed: the observed peak annual-mean overturning carries 0.51 $\mathrm{m\,s^{-1}}$ as a half-column branch velocity, 1.31 $\mathrm{m\,s^{-1}}$ as a 195 hPa slab, and 1.07 $\mathrm{m\,s^{-1}}$ as a 238 hPa slab. The model's v is a slab velocity, so model-versus-ERA5 comparisons of max $|v|$ belong against the last two. The 0.49 $\mathrm{m\,s^{-1}}$ quoted in my 2026-07-31 notes was the half-column number and should not have been compared with the model.

6 The moisture coefficient

There are three different things one can mean by a, and they answer three different questions.

The mass partition is $a = W_{\mathrm{lower}}/W_{\mathrm{total}}$ at a chosen interface. It answers "where is the water". At the half-mass level $p_s/2$ it is 0.9501 over $|\varphi| \le 20°$ (ERA5 1979–2023), and it barely moves: 0.9477 within 10°, 0.9506 within 30°, 0.9501 globally. Its seasonal cycle in the tropical mean spans 0.0012, from 0.9496 in March to 0.9508 in October, because the hemispheres cancel; at a fixed latitude the swing is twenty times larger, 0.939 to 0.964 at 15°N with the maximum in local winter. Across 45 annual means it spans 0.9466 to 0.9529 with a trend of -0.0011 per decade. Every one of those variations is dwarfed by the choice of interface: 0.982 at 400 hPa, 0.950 at 500, 0.883 at 600, 0.772 at 700.

The flux-matched coefficient is the regression slope of the observed column moisture flux on the model's transport basis vW. It answers "what coefficient reproduces the observed transport", which is what the model needs, and it absorbs the correlation between $[v]$ and $[q]$ within each branch. It depends on the reading of v, and that is the whole content of Table 2.

The literal two-slab partition is $(W_{\text{lower slab}} - W_{\text{upper slab}})/W$ for slabs of depth dp anchored at the ground and at the model's tropopause. It answers "what would two genuinely uniform slabs carry". At $dp = 196$ hPa it is 0.588 against a flux-matched 0.547, so the uniform-slab idealization would over-transport by 7%. The sign of that correction is opposite to the half-column case, where the flux-matched 1.401 exceeded the literal 0.900 by 56%, because the half-column lower branch is 500 hPa deep and its water sits at the very bottom where $|v|$ is largest, whereas a 196 hPa slab is thinner than the real lower branch and reaches less far into dry air.

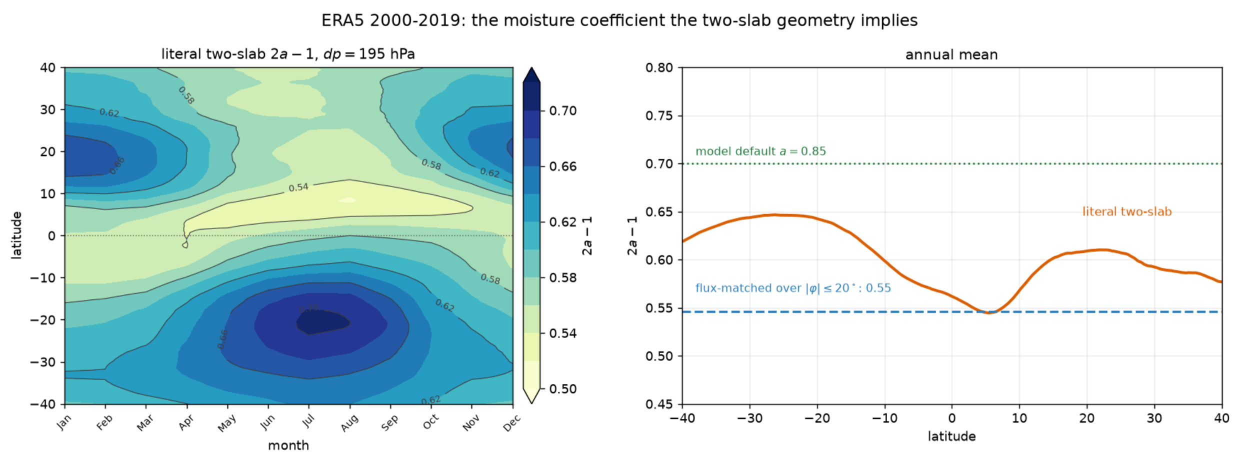

Figure 3: The literal two-slab coefficient at $dp = 195$ hPa across latitude and season (left), and its annual mean against the flux-matched value and the model's default (right). The seasonal structure is the monsoon signature: the lower slab holds a larger share of the column in the winter subtropics, where subsidence and the trade inversion trap water low.

7 How the time mean is taken, and why it is the largest uncertainty

The regressions above pool every month and latitude, so they weight by transport and are dominated by the strong monsoon months. The model is steady and has no seasonal cycle, so a time-mean circulation and its time-mean transport are the closer counterpart. Four defensible choices give Table 3.

Table 3: The moisture coefficient and the regime boundary under four ways of taking the time mean, all at $dp = 195$ hPa, ERA5 2000–2019, $|\varphi| \le 20°$. $\hat{H}/\hat{S}$ is at the observed $W = 39.6\ \mathrm{kg\,m^{-2}}$.

time mean	$\hat{S}$ ($\mathrm{J\,m^{-2}}$)	$2a-1$	a	W^* ($\mathrm{kg\,m^{-2}}$)	$\hat{H}/\hat{S}$
every month, transport-weighted	7.746×10^7	0.546	0.773	56.8	0.303
annual mean, $\bar{v}\,\bar{W}$	7.581×10^7	0.589	0.795	51.5	0.267
equinox months, $\bar{v}\,\bar{W}$	7.579×10^7	0.589	0.794	51.5	0.259
annual mean, $\overline{vW}$	7.581×10^7	0.683	0.842	44.4	0.107

The spread is seasonal covariance. The last row keeps the correlation, within the seasonal cycle,

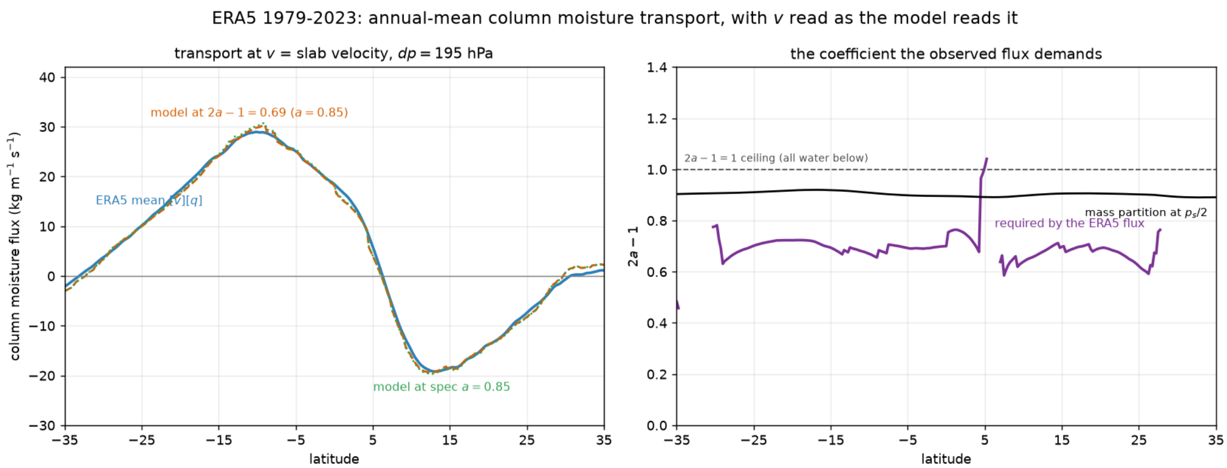

Figure 4: Annual-mean column moisture transport, with v read as the model reads it. Left: the observed flux against the model's structure at the flux-matched coefficient and at the spec value $a = 0.85$. Right: the coefficient the observed flux demands, now comfortably below its own ceiling. Under the half-column reading this curve sat at 1.6 to 1.8, above the ceiling, which was the visible symptom of the mismatch.

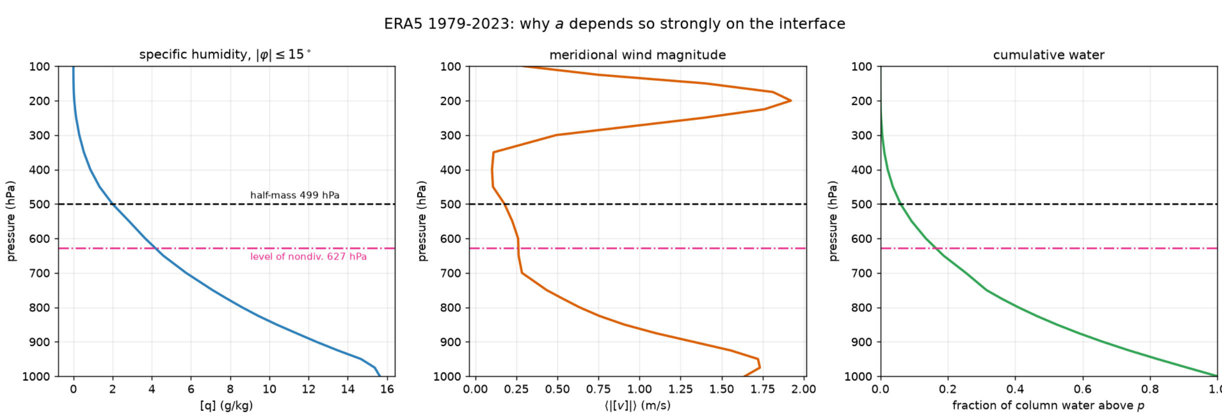

Figure 5: Why a depends so strongly on the interface: the deep-tropical $[q]$ and $|[v]|$ profiles and the cumulative column water. Half the column mass sits above 499 hPa; about 5% of the column water does.

between a strong lower branch and a moist column, so its transport basis is the largest and the coefficient it demands is the smallest per unit transport [plausible]. The second row removes that correlation by multiplying the two annual means. The equinox row, restricted to March, April, September and October, is the closest analogue of the model, which is steady, hemispherically symmetric and has no solstitial cell to average away. That it lands on the annual-mean value to three decimal places is reassuring: the worry that the annual-mean tropical overturning is a small residual of two large opposing cells, and therefore a bad target, does not bite here.

The consequence is that the observations do not settle the sign of the model's gross moist stability at $W_c = 50$. W^* from 44.4 to 56.8 $\mathrm{kg\,m^{-2}}$ straddles the 50.7 $\mathrm{kg\,m^{-2}}$ quiescent column. What the observations do settle is comparative, and that is enough to act on: $a = 0.85$ lies at or just beyond the strong-transport end of the range, and gives the lowest W^* of any of them.

Recommendation: take a from the equinox row, $a \approx 0.79$, and carry 0.77 to 0.84 as its uncertainty.

8 The gross stabilities and their error budget

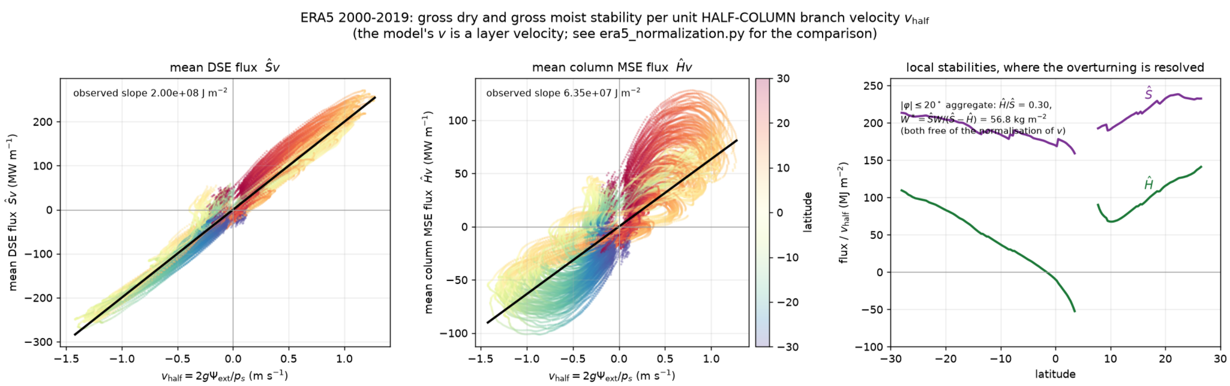

Figure 6: The regressions the stabilities come from, per unit half-column branch velocity. (a) and (b) are the scatters themselves, coloured by latitude, so a poor fit or a latitude-dependent slope would be visible rather than hidden inside one number. (c) is the same quantity resolved in latitude instead of aggregated. The local $\hat{H}$ goes negative within a few degrees of the equator; see the caveat in the text.

Both stabilities are regression slopes against the same velocity, so neither requires choosing a branch interface. Over four latitude bands (ERA5 2000–2019, half-column normalization, mass-adjusted):

band	$\hat{S}_{\mathrm{eff}}$	$\hat{S}_{\mathrm{bulk}}$	$\hat{H}_{\mathrm{eff}}$	W ($\mathrm{kg\,m^{-2}}$)	$\hat{H}/\hat{S}$
$\lvert\varphi\rvert \leq 10^\circ$	1.909×10^8	2.287×10^8	4.93×10^7	44.5	0.258
$\lvert\varphi\rvert \leq 15^\circ$	1.946×10^8	2.283×10^8	5.46×10^7	42.2	0.281
$\lvert\varphi\rvert \leq 20^\circ$	1.978×10^8	2.283×10^8	5.99×10^7	39.6	0.303
$\lvert\varphi\rvert \leq 30^\circ$	1.997×10^8	2.301×10^8	6.35×10^7	35.1	0.318

Two caveats carry into everything above.

The barotropic mass adjustment is the dominant uncertainty on $\hat{S}$. ERA5's zonal-mean $[v]$ on pressure levels carries a spurious net column mass flux, which leaves $|\Psi(p_s)|$ averaging 232 and peaking at 780 $\mathrm{kg\,m^{-1}\,s^{-1}}$. Removing the mass-weighted column mean of $[v]$ first changes $\hat{S}$ by -33.8%, from 2.989×10^8 to 1.978×10^8, while the peak $|\Psi_{\text{ext}}|$ moves by only -10.2% and the peak $|v|$ by -10.3%. $\hat{S}$ is the most sensitive of the three because the transported quantity s has a mean near 330 $\mathrm{kJ\,kg^{-1}}$, so a small spurious barotropic $[v]$ fabricates a large spurious flux, while Ψ and v feel only the mass flux itself. Treat 34% as the floor on the uncertainty in $\hat{S}$ until the mass-consistent ERA5 product is used. Moisture is far more tolerant of the same error because q has a small mean, which is why the D route in Section 11 is robust and the energy route is not.

The local $\hat{H}$ is not uniformly positive. Panel (c) of Fig. 6 shows $\hat{H}$ falling through zero within about 8° of the equator, reaching roughly -50 $\mathrm{MJ\,m^{-2}}$ just south of it, and rising to $+140$ $\mathrm{MJ\,m^{-2}}$ by 25°N. The band aggregate is dominated by the large-$|v|$ subtropical points. The equatorial values are poorly conditioned, since v passes through zero there and the local ratio is a ratio of two small numbers, so do not quote -50 as a firm number. This is the observed counterpart of the slightly negative gross moist stability that Spencer said might still be of interest.

9 The observed fluxes, mean against eddy

The gross stabilities of Section 8 are properties of the mean circulation alone. Fig. 7 puts that mean transport next to the eddy transport it competes with, for moisture and for moist static energy, with moisture converted to energy units by multiplying by L_v so the two share an axis.

Both splits use the same route. The mean part comes straight from the zonal means, $(1/g)\int[v][x]\,dp$. The total comes from the column budget, whose source is a stored surface or top-of-atmosphere field and so contains every eddy the reanalysis has, stationary and transient. The eddy part is the difference. Zonal averaging destroys the covariance that would let the eddy flux be formed directly, which is why the budget route is needed at all. For moisture the source is $E - P$; for energy it is the net downward flux at the top of the atmosphere minus the net downward flux at the surface, radiative and turbulent.

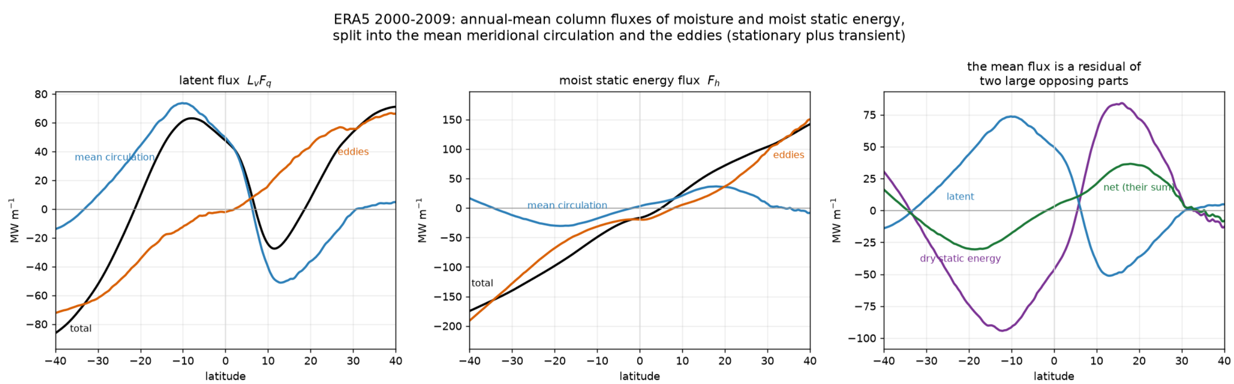

Figure 7: Annual-mean column fluxes, positive northward, per unit length along a latitude circle. (a) Latent flux L_vF_q: the mean circulation carries moisture equatorward into the ITCZ in both hemispheres while the eddies carry it poleward. (b) Moist static energy: the mean flux is small and the eddies dominate poleward of about 20°. (c) The mean MSE flux decomposed. It is the residual of a dry-static-energy flux and an opposing latent flux, each two to three times larger than their sum.

Three things are worth reading off it.

The mean MSE flux is a residual, and a fairly small one. At 15°N the mean circulation carries +83.1 $\mathrm{MW\,m^{-1}}$ of dry static energy poleward against −49.0 $\mathrm{MW\,m^{-1}}$ of latent energy equatorward, netting +34.1. At 10°N it is +61.3 against −40.4, netting +20.9. That ratio, roughly $1 - W/W^*$, is the observational content of $\hat{H}/\hat{S} = 0.30$ from Section 2, seen in the flux rather than in the coefficient.

The sign of the mean MSE flux is poleward in the subtropics of both hemispheres, +35.4 $\mathrm{MW\,m^{-1}}$ at 20°N and −30.0 at 20°S. The mean circulation exports moist static energy out of the deep tropics, which is the positive-gross-moist-stability behaviour, and it is what the model at $a = 0.85$ and $W_c = 50$ failed to reproduce (Table 4, where that run has the mean flux at +104.7 $\mathrm{MW\,m^{-1}}$ against an eddy flux of 106.2 in near-cancellation).

The eddies dominate outside the deep tropics. The eddy MSE flux passes the mean near 18°N and reaches +96.6 $\mathrm{MW\,m^{-1}}$ at 30°N against a mean of +7.1. The model has no representation of that transport for energy: its only eddy term is the moisture diffusion $-D\,\partial_y W$.

Two caveats. The moisture budget closes well, leaving +4.2 and −1.4 $\mathrm{MW\,m^{-1}}$ at 85°N and S where the flux must vanish, against a tropical signal near 60. The energy budget closes worse, leaving +31.0 and −27.1 $\mathrm{MW\,m^{-1}}$ against a peak near 140, about 20%. That is a latitude-dependent bias in ERA5's energy budget which removing the global mean cannot fix, so the eddy MSE curve is the least certain thing in this figure. For scale, the total MSE flux at 15°N is 51.9 $\mathrm{MW\,m^{-1}}$, which multiplied around the latitude circle is 2.01 PW, a reasonable atmospheric energy transport there.

10 Does the ERA5-consistent a remove the aggregated regime?

The prediction is direct: hold everything else fixed, lower a from 0.85 to the ERA5-consistent end, and a $W_c = 50$ run should stop being negative-stability. One run tests it. The control is the $W_c = 40$ run of 2026-07-31, which reaches positive stability the other way, by lowering the column instead of raising the crossover.

Table 4: The V2 test. W^* and W_q are the model's own values. "Bands" counts contiguous regions with $P > 0.5\ \mathrm{mm\,day^{-1}}$; "dry" is the fraction of the domain where P is identically zero. Mean and eddy MSE fluxes are peak values in $\mathrm{MW\,m^{-1}}$.

run	W^* ($\mathrm{kg\,m^{-2}}$)	W_q ($\mathrm{kg\,m^{-2}}$)	min $\hat{H}$ ($\mathrm{MJ\,m^{-2}}$)	bands	dry	$P_{\max}$ ($\mathrm{mm\,day^{-1}}$)	mean flux ($\mathrm{MW\,m^{-1}}$)	eddy flux ($\mathrm{MW\,m^{-1}}$)
$a = 0.85$, $W_c = 50$	44.2	50.66	−32.8	5	0.36	77.8	+104.7	106.2
$a = 0.77$, $W_c = 50$	57.4	50.66	+8.3	1	0.00	7.4	+17.0	1.0
$a = 0.85$, $W_c = 40$	44.2	40.66	+4.8	1	0.00	9.0	+14.5	1.5

The prediction holds. At $a = 0.77$ the five precipitating bands collapse to one, the dry gap closes, the peak precipitation falls from 78 to 7.4 $\mathrm{mm\,day^{-1}}$, the jet weakens from 73.8 to 58.7 $\mathrm{m\,s^{-1}}$ while its deepest easterly notch shallows from −45.5 to −16.6 $\mathrm{m\,s^{-1}}$ and moves poleward from 7.48 to 9.53 Mm, and the mean MSE flux returns to a modest poleward +17 $\mathrm{MW\,m^{-1}}$ with the eddy flux a 1.0 $\mathrm{MW\,m^{-1}}$ bit player, against +104.7 and 106.2 in near-cancellation before. Every one of those numbers lands next to the $W_c = 40$ control rather than next to the run it was changed from. The aggregated regime was the negative gross moist stability, and the negative gross moist stability was the calibration error.

That said, the resulting solution is still not a good match to the observed circulation. Its jet peaks

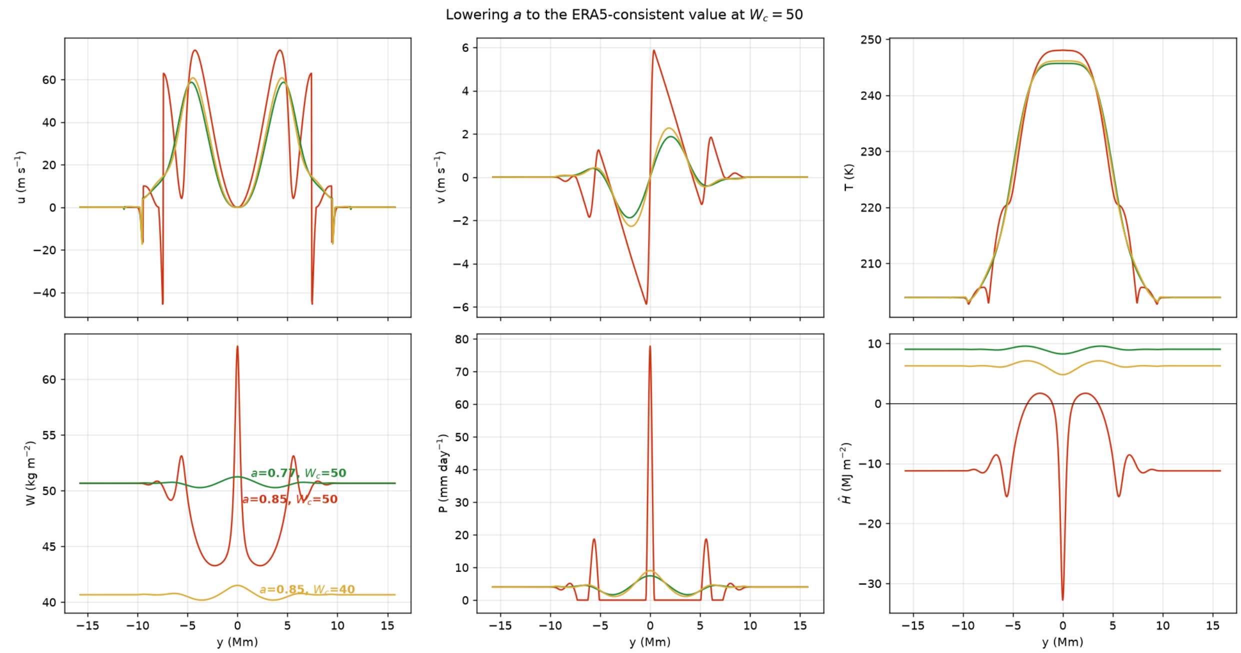

Figure 8: Equilibrium profiles at $ny = 801$, $dt = 30$ s, converged under the slow-drift gate. Red: the 2026-07-31 configuration, $a = 0.85$ and $W_c = 50$. Green: the same run with $a = 0.77$. Yellow: the $W_c = 40$ control at $a = 0.85$. Green and yellow are close in every field; red is the aggregated regime.

at 58.7 $\mathrm{m\,s^{-1}}$ against an observed 200 hPa zonal-mean maximum of 31.0 $\mathrm{m\,s^{-1}}$, and its meridional temperature contrast is 1.4 times the observed one at 16° and 24°, 1.6 times at 33° (Table 5). Fixing the moisture coefficient removes a pathology; it does not make V2 realistic, and the overshoot of the jet is common to every V2 configuration run so far.

11 The other parameters

Eddy diffusivity D. Zonal-mean fields cannot form the eddy moisture flux directly, because zonal averaging has already destroyed the covariance. The steady column budget routes around that: the total flux follows from $E - P$ by meridional integration, and subtracting the mean flux $\int [v][q]\,dp/g$ leaves the eddy flux, stationary and transient together. Then $D = -F_{\text{eddy}}/(\partial_y W)$. Over ERA5 2000–2009 the median over $|\varphi| \leq 25°$ is 7.2×10^5 $\mathrm{m^2\,s^{-1}}$ with quartiles 4.1×10^5 to 1.8×10^6, but the latitude structure is the real result: D rises from 2.7×10^5 at 5°S and 5–6 $\times 10^5$ near 10° to 2.0×10^6 at 20°S, 2.8×10^6 at 20°N and 4.4×10^6 at 25°N. An order of magnitude separates the deep tropics from the subtropics. That reconciles the model default $D = 10^6$, the earlier aggregate near 7×10^5, and the postdoc's lat-lon tropical estimate of 1 to 2×10^5: the model's default is a subtropical value applied to the deep tropics, and the residual gap to the lat-lon estimate is plausibly the stationary eddies that the zonal-mean route lumps into the same term. The budget closes: the total flux reaches only 1.7 and -0.5 $\mathrm{kg\,m^{-1}\,s^{-1}}$ at 85°N and S, where it must vanish, once the global mean $E - P$ imbalance is removed first.

Evaporation E_0. ERA5 gives 4.58×10^{-5} within 10°, 4.79×10^{-5} within 20° and 4.62×10^{-5} within 30° $\mathrm{kg\,m^{-2}\,s^{-1}}$. The model default is 4.6×10^{-5}. No change is needed.

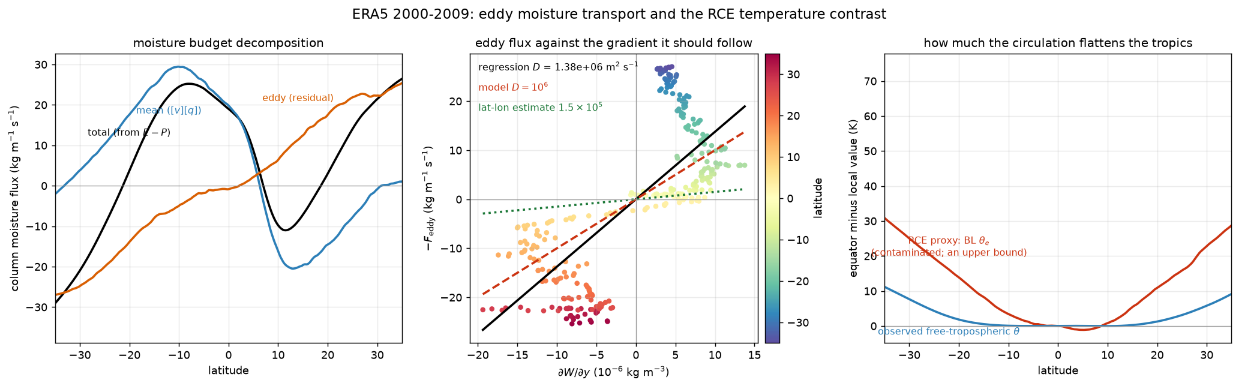

Figure 9: (a) The column moisture budget decomposed into its total, mean and eddy parts. (b) The eddy flux against the gradient it should follow, with the regression slope, the model default and the postdoc's lat-lon estimate. (c) The boundary-layer θ_e contrast against the observed free-tropospheric θ contrast; the former is a contaminated proxy for the radiative-convective target, so it bounds rather than determines it.

The radiative-convective temperature contrast. Spencer's position of 2026-08-01, which this report adopts: the relaxation target in V2 should be radiative-convective equilibrium rather than radiative equilibrium, and he does not know how he convinced himself otherwise when the spec was written. He is also right that the target is not observable. A column-by-column RCE free troposphere sits on the moist adiabat through its own boundary layer, so its meridional structure would follow the boundary-layer θ_e, whose equator-to-24° contrast is 16.6 K against 3.2 K for the observed free troposphere, a factor of 5.2. But boundary-layer θ_e in a reanalysis already carries the circulation's influence, and a true latitude-by-latitude RCE has time-mean $v = w = 0$ by definition, so it is not a state of the real atmosphere at all. Those two numbers therefore bound $\Delta\theta^{\mathrm{rad}}$ rather than determine it. Treat it as a tuning parameter with ERA5 setting the limits.

β and the latitude mapping. The model's $\beta = 2 \times 10^{-11}\ \mathrm{m^{-1}\,s^{-1}}$ equals Earth's $2\Omega\cos\varphi/a$ at $\varphi = 29.1°$. Following Spencer's suggestion to compute a proper Hadley-range average, $\langle\beta\rangle = 2\Omega\int\cos^2\varphi\,d\varphi \,/\, \left(a\int\cos\varphi\,d\varphi\right)$ over 0 to 30° gives 2.190×10^{-11}, 9.5% above the model's value; the equatorial value is 2.289×10^{-11} and the value at 15° is 2.211×10^{-11}. For turning model y into a latitude, the Coriolis mapping $f = \beta y = 2\Omega\sin\varphi$ is the one to use, since it preserves the dynamically relevant quantity: $y = 2$, 3 and 4 Mm map to 16°, 24° and 33°. Arc length on Earth's radius gives 13°, 27° and 54° instead, and the difference at large y is substantial.

12 Model against ERA5

The ERA5-consistent run is the closest of the three to the observations in every column, and it is still too steep by a factor of 1.4 to 1.6 in the temperature contrast and too strong by a factor of 1.9 in the jet. The jet overshoot is shared by every V2 configuration and predates this calibration; the 2026-07-30 attribution chain traced it to the radiative retarget steepening the relaxation target, a comparison not re-run for this report. It is a separate problem from the moisture coefficient.

The peak $|v|$ comparison in Table 5 replaces the one in my 2026-07-31 notes, which mixed conventions by comparing the model's slab velocity against ERA5's half-column velocity.

Table 5: Equilibrium model states against ERA5. Temperature contrasts are $\theta(0)-\theta(y)$ in potential temperature, evaluated at the Coriolis-mapped latitudes. Peak $|v|$ for ERA5 is quoted as a slab velocity at $dp = 195$ and 238 hPa, the two depths of Section 5, because that is the quantity the model's v is; the half-column value is 0.51 $\mathrm{m\,s^{-1}}$.

	$\Delta\theta$ at 16°	at 24°	at 33°	jet ($\mathrm{m\,s^{-1}}$)	peak $|v|$ ($\mathrm{m\,s^{-1}}$)
ERA5	0.56	3.21	8.75	31.0	1.07 to 1.31
V2, $a = 0.85$, $W_c = 50$	1.69	7.69	21.33	73.8	5.86
V2, $a = 0.77$, $W_c = 50$	0.76	4.49	14.20	58.7	1.88
V2, $a = 0.85$, $W_c = 40$	0.90	5.13	15.72	60.8	2.27

13 Outstanding questions

Things ERA5 could settle but has not, for want of data. Five ERA5 directories that would remove the largest uncertainties here are present but empty: `col_wv_flux_north`, `col_mse_flux_north_st_eddy`, `col_dse_flux_north_st_eddy`, `div_col_wv_flux` and `tend_col_wv`. The first three would give the column fluxes directly from ERA5's own mass-consistent, submonthly accounting, instead of reconstructing them from monthly zonal means. That would retire the 34% mass-adjustment uncertainty on $\hat{S}$ (Section 8), separate the stationary from the transient eddies in D, and remove the aggregation ambiguity of Section 7, which is currently the single largest unknown in the calibration. Downloading them is the highest-value next step by a wide margin.

Which time mean is the right calibration target. Section 7 argues for the equinox product-of-means on the grounds that the model is steady and symmetric, and the equinox and annual-mean routes agree. The mean-of-products route, which keeps the seasonal covariance, gives a coefficient 16% larger and puts W^* exactly at the model's default value. Deciding this properly needs a statement of what the steady model is meant to represent, which is a modeling question rather than an observational one.

Parameters still to calibrate. Δ_z and H directly from ERA5, by taking θ at the lapse-rate tropopause minus θ at the surface and the tropopause height, which would check whether 60 K and 16 km hold up; τ from the clear-sky radiative cooling fields `rlntcs`, `rsntcs` and `olr`, all present on disk; W_c and τ_c from regressing P on W, noting that monthly zonal means give only a smoothed pickup curve and so bound rather than fit the pair; y_1 and y_0; and optionally v_d from the zonal-momentum-budget residual. θ_{00} is cosmetic in V2 and needs no calibration.

Whether d should be re-specified in pressure. The model carries d in metres and combines it with H in metres. Everything in Section 5 says the meaningful parameter is the layer's pressure depth. The two are interchangeable at fixed $\hat{S}$, so this is a question of whether to make the model's own statement of its assumptions honest rather than a question of changing any result.

The V2 jet overshoot. The three configurations of Section 10 produce jets 1.9 to 2.4 times the observed 200 hPa maximum, and the $\Delta\theta^{\mathrm{rad}}$ ladder of 2026-07-30 reaches higher still. In that same ladder, latent heating steepened the model's meridional temperature gradient rather than flattening it, which is the opposite of the rationale given in the spec; that comparison has not been re-run since. The overshoot is now the largest remaining model-observation discrepancy, and it is untouched by this calibration.

The slightly negative gross moist stability case. Spencer flagged this as possibly still of interest. The observations support it locally: $\hat{H}$ is negative within roughly 8° of the equator in

ERA5 (Section 8), even though the tropical aggregate is firmly positive at $\hat{H}/\hat{S} = 0.30$. A model configuration with W_q just above W^* would be the analogue, and it is now locatable in closed form.

A Equations and parameter values

Transcribed from `ss09/sw_model.py` (`du_dt`, `dv_dt`, `dtheta_dt`, `moisture_transport_tendency`) and cross-checked against `SCIENCE.md` §2.1–2.3 and §3.1. θ_{rad} is the radiative relaxation target, $\Lambda = L_v/C$, $T = \Pi\theta$ with $\Pi = 1/1.6$, and $\mathcal{H}$ is the Heaviside step with $\mathcal{H}(0) = 1/2$:

$$\partial_t u = \beta y v - v\,\partial_y u - \mathcal{H}(G)\,(\partial_y v)\,u - \epsilon_u u - v_d\,\mathcal{H}(u)\,\mathrm{sgn}(y)\,\partial_y u, \tag{12}$$

$$2\,\partial_t v = -\beta y u - \frac{gH}{T_0}\,\partial_y T + \kappa_v \partial_y^2 v, \tag{13}$$

$$\partial_t\theta + \frac{\delta\,\Delta_z}{H}\,\partial_y v = \frac{\theta_{\mathrm{rad}} - \theta}{\tau} + \kappa_\theta \partial_y^2\theta + \Lambda P, \tag{14}$$

$$\partial_t W + \partial_y[-(2a-1)vW - D\,\partial_y W] = E_0 - P, \qquad P = \frac{(W - W_c)^+}{\tau_c}. \tag{15}$$

Four points where these differ from what a reader might expect, each of which I had wrong in the first version of this report:

- The eddy momentum flux divergence in (12) is an *advection* at the poleward velocity $v_d\,\mathrm{sgn}(y)$, gated on westerlies. It is not the divergence of a flux $v_d\mathcal{H}(u)u$. Because $\mathcal{S} \propto \mathrm{sgn}(y)\,\partial_y u$ it is up-gradient at a wind maximum, which is what makes the jet flanks the delicate region of the model.
- The pressure-gradient term in (13) is written directly in temperature, $-(gH/T_0)\partial_y T$, with $T_0 = 300$ K a fixed reference. There is no geopotential in the model. Using θ in place of T here was one of the two errors fixed by the SS13 corrigendum.
- The vertical-advection gate $\mathcal{H}(G)$ sits on the *momentum* equation only. $G = \theta_E - \theta$ for dry and V1 runs, $G = \partial_y v$ under the `kinematic` gate that V2 uses. The thermodynamic equation (14) is ungated.
- δ is the model's layer depth, called d in the body of this report and `delta` in the code. κ_θ is zero by default.

Column constants (`ss09/moist_constants.py`): $C = c_p\Pi\,\Delta p/g = 5.164 \times 10^6\ \mathrm{J\,m^{-2}\,K^{-1}}$, $L_v = 2.5 \times 10^6\ \mathrm{J\,kg^{-1}}$, $\Lambda = L_v/C = 0.4843$ K per $\mathrm{kg\,m^{-2}\,s^{-1}}$, $\hat{S} = Cd\Delta_z/H = 7.746 \times 10^7\ \mathrm{J\,m^{-2}}$. The identity $C\Lambda = L_v$ is what makes precipitation cancel from the column MSE budget.

Parameter values used above: $\delta = 4$ km, $\Delta_z = 60$ K, $H = 16$ km, $T_0 = 300$ K, $g = 9.81\ \mathrm{m\,s^{-2}}$, $\beta = 2\times10^{-11}\ \mathrm{m^{-1}\,s^{-1}}$, $\tau = 3.197\times10^6$ s, $v_d = 2.5\ \mathrm{m\,s^{-1}}$, $a = 0.85$, $D = 10^6\ \mathrm{m^2\,s^{-1}}$, $W_c = 50\ \mathrm{kg\,m^{-2}}$, $\tau_c = 14400$ s, $E_0 = 4.6 \times 10^{-5}\ \mathrm{kg\,m^{-2}\,s^{-1}}$, $\Delta\theta^{\mathrm{rad}} = 75$ K, $y_1 = 9439$ km. Quiescent column $W_q = W_c + \tau_c E_0 = 50.66\ \mathrm{kg\,m^{-2}}$.

One inconsistency to be aware of when reading numbers across this report: the model uses $g = 9.81$ while the ERA5 scripts use $g = 9.80665$, so $\hat{S}$ and W^* computed on the two sides differ in their fourth digit. That is why Table 1 gives $W^* = 44.3\ \mathrm{kg\,m^{-2}}$ from the ERA5-side calculation and Table 4 gives 44.2 from the model runs' own attributes. It changes nothing.

B Parameter status

status	parameters
calibrated here	$\hat{S}$, $\hat{H}$, a and $2a-1$, E_0, D, W^*
calibrated with a stated caveat	$\Delta\theta$ (contaminated proxy), β (a choice of reference latitude)
calibratable, not yet done	Δ_z, H, τ, y_1, y_0, (W_c, τ_c), v_d
over-determined	d: two slab parameters against three observables, 13% residual
not observable	$\Delta\theta^{\mathrm{rad}}$ (needs offline RCE), θ_{00} (cosmetic)
purely numerical	ϵ_u (regularization), κ_v

The organizing principle behind this table: calibrate the combinations the equations use, not the individual symbols. Calibrating a on its own is what pushed the gross moist stability negative, because the symbols only mean something once the normalization of v is fixed. W^* and $\hat{H}/\hat{S}$ are the combinations that survive that ambiguity, and they are where the comparison with observations belongs.

C Data and scripts

ERA5 monthly zonal means at `~/Dropbox/data/ecmwf/era5/monthly/`, symlinked in the repository as `era5-data/`. Fields used: `hus`, `cwv`, `ps`, `va`, `ua`, `ta`, `zg`, `evap`, `pr`. Precipitation is the only latitude-longitude file (4 GB) and is streamed month by month, since dask is not installed in this environment. Year ranges differ by script for cost reasons and are stated with each result: 1979–2023 for the mass partition, 2000–2019 for the stabilities and the normalization, 2000–2009 for the moisture budget.

Scripts, all run as `python scripts/<name>.py` from the repository root:

script	what it produces
`era5_normalization.py`	Sections 2, 4, 7; Figs. 1, 3
`era5_vertical_structure.py`	Section 5; Fig. 2
`era5_stability_calibration.py`	Section 8; Fig. 6
`era5_a_calibration.py`	the mass partition of Section 6; writes the cache
`era5_a_calibration_figures.py`	Figs. 4, 5
`era5_moisture_budget.py`	Section 11; Fig. 9
`era5_flux_decomposition.py`	Section 9; Fig. 7
`era5_calibration_v2_check.py`	Section 10; Fig. 8, Table 5

The V2 test run of Section 10 is `model_output/moist_v2/wc50_a077/`, produced by

```
run-sw-model --stop-at-steady-state --slow-drift-gate
--ny 801 --dt 30 --backend numba
--enable-moisture --enable-latent-heating
--cwv-frac 0.77 --w-crit 50
```

converging at day 1392, alongside the existing `model_output/moist_v2/m4_v2_dyr75/` and `model_output/moist_v2/wc40/`.

D Two verification checks worth recording

The identity behind (1) is checked numerically: the model's $-(2a-1)vW$ and the two-layer bulk flux evaluated at the half-mass interface agree to 2.8×10^{-14} $\mathrm{kg\,m^{-1}\,s^{-1}}$, which is $5 \times 10^{-14}\%$ of the peak flux. The two expressions are the same algebra, and the code agrees.

The pressure-level column integral of $[q]$ runs 0.56 $\mathrm{kg\,m^{-2}}$ below ERA5's own total column water vapour in the tropics, a mean relative error of 1.95%, with the largest discrepancies over Tibet and the Sahara where zonal asymmetry in surface pressure is largest. Piecewise-constant against trapezoidal integration changes a by 0.0006, and removing the below-ground mask entirely changes it by 0.0007. Spencer's call on 2026-08-01 was that an error of this size is not a concern.